

Grundlagen Digitaler Medien

S3D Video – Content Creation

Dozent: Prof. Dr.-Ing. Sebastian Knorr

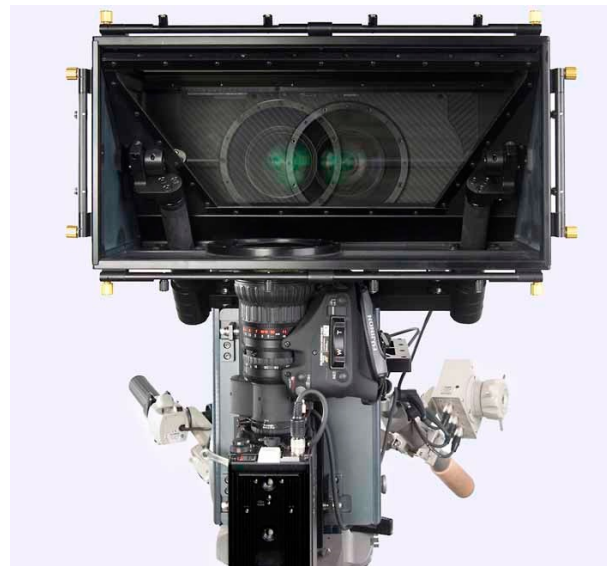
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Lernziele

- Verständnis für natives Stereo-3D
- Verständnis für die 2D-3D Konvertierung
- Verständnis für die Verarbeitungsschritte der S3D Erstellung

Natives S3D



Know How in Native 3D Productions

3D Storytelling

- Directors
- Stereographers



3D Shooting

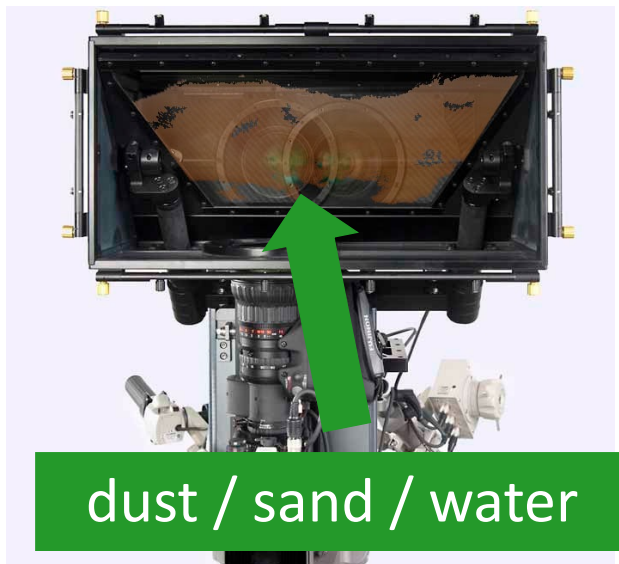
- Camera operators
- Stereographers



3D Post-production

-
- S3D operators
 - S3D supervisors
 - Stereographers
-
- | | | |
|---|---|--|
| <ul style="list-style-type: none">• 2 scripts
(screenplay and
depth script) | <ul style="list-style-type: none">• Camera equipment• Displays and screens
for pre-viz | <ul style="list-style-type: none">• Mistika• Nuke & Occula• Speedgrade DI• etc. |
|---|---|--|

3D Shooting Conditions



Mirror Rig



3D Post-Production



SGO Mistika – Stereo Finishing



Binocular rivalry →
Visual discomfort

Color Mismatch



Color Mismatch



Geometrical Misalignments



Binocular rivalry in 3D projects

1. Vertical misalignment



mostly occurs in/ how to fix it:

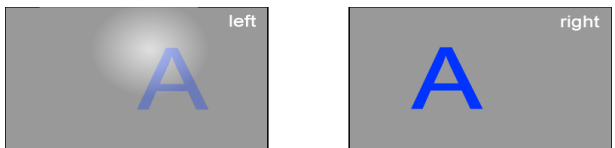
→ 3D live production:
geometry realignment in post

2. Luminance



→ 3D live production:
color adjustment in post

3. Reflections, Flares, Polarization



→ 3D live production:
significant post-production work

Binocular rivalry in 3D projects

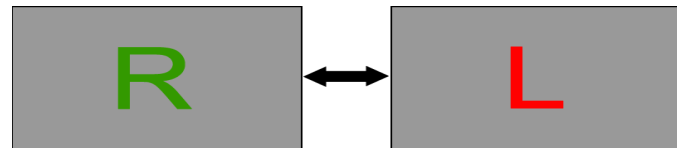
4. Contamination



5. Depth of field



6. Pseudo 3D



mostly occurs in/ how to fix it:

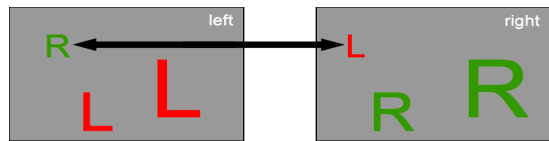
→ 3D live production:
dust removal in post

→ 3D live production:
significant post-production work

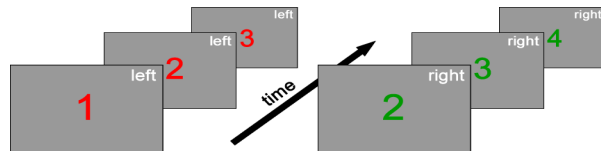
→ 3D live production:
swap left and right images

Binocular rivalry in 3D projects

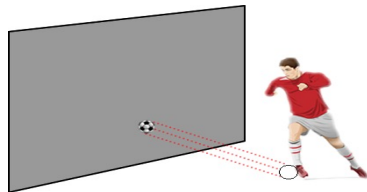
7. Partial pseudo 3D



8. Synchronization



9. Depth mismatch



mostly occurs in/ how to fix it:

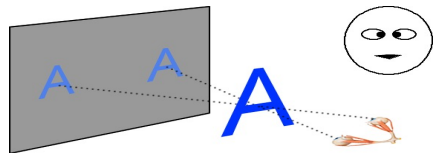
→ realtime conversion:
cannot be fixed entirely

→ 3D live production:
significant post-production work

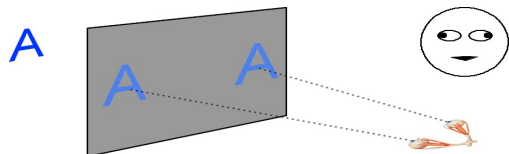
→ realtime conversion:
complex image processing

Binocular rivalry in 3D projects

10. Hyperconvergence



11. Hyperdivergence



12. Visual mismatch



mostly occurs in/ how to fix it:

→ 3D live production:
reduce negative parallax

→ 3D live production:
reduce positive parallax

→ conversion:
fix 3D composition

Binocular rivalry in 3D projects

13. Ghosting

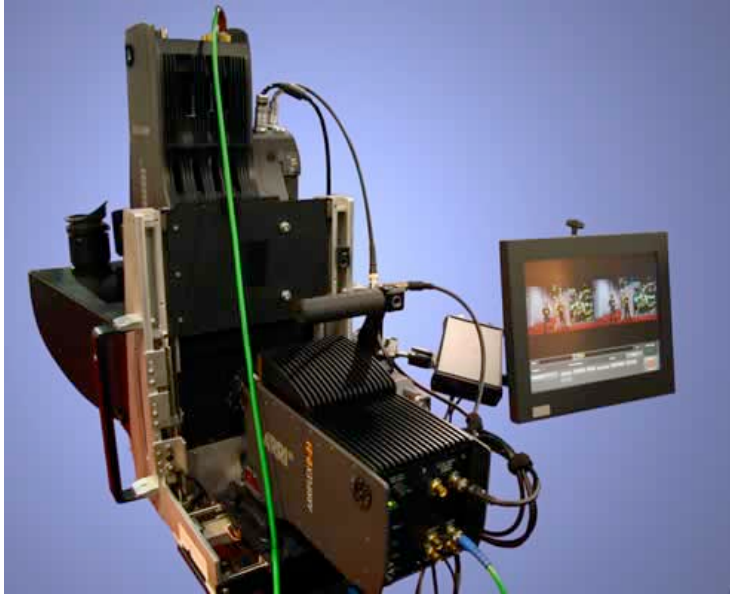


mostly occurs in/ how to fix it:

→ all productions (displaying issue):

- reduce contrast
- reduce depth budget
- change convergence

Realtime Analysis and Correction



Binocular Rivalry in 3D Projects

- Reasons for visual discomfort:
 - Conflicts of depth cues (monocular vs. binocular) and binocular rivalry
- When all cues line up, stereo is excellent
- In case of a conflict of cues, stereo breaks down
- Not all cues have to be “perfect”, but it helps



2D-to-S3D Conversion

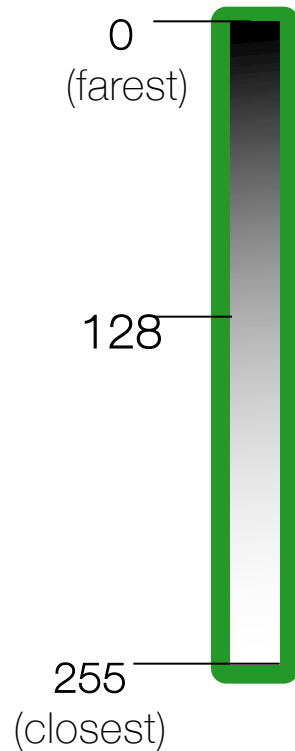
Recovering of Depth Information



original image



depth map

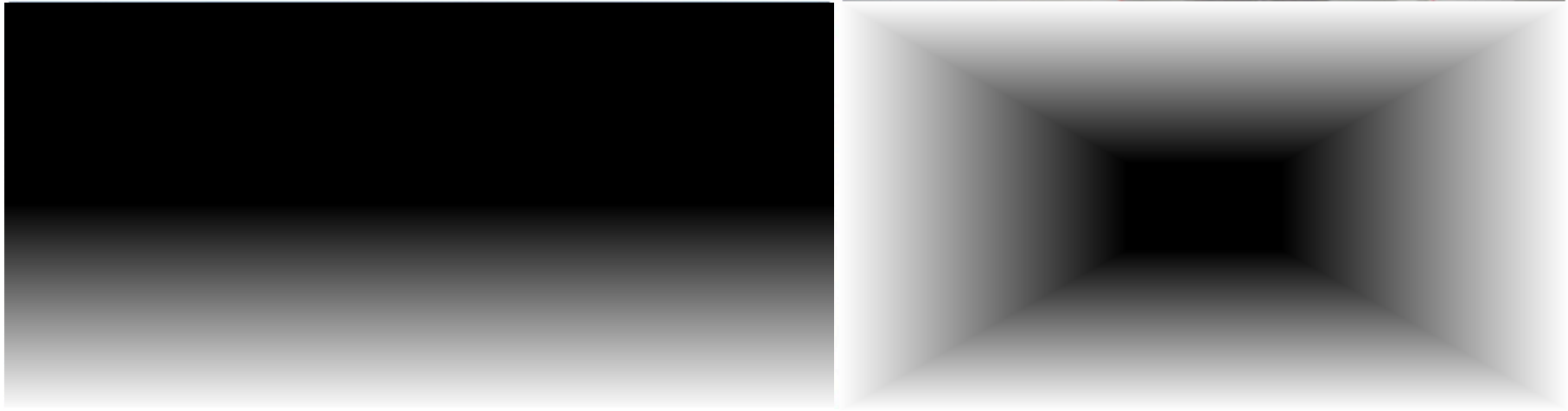


Simple 2D to 3D Conversion



perspective cues: vanishing lines

Simple 2D to 3D Conversion



perspective cues: vanishing lines

Hybrid Conversion Techniques



original



color

Hybrid Conversion Techniques



original



color + gradient

Automatic and Realtime Conversion



error-prone depth maps: conflicts of depth cues, pseudo 3D

→ visual discomfort

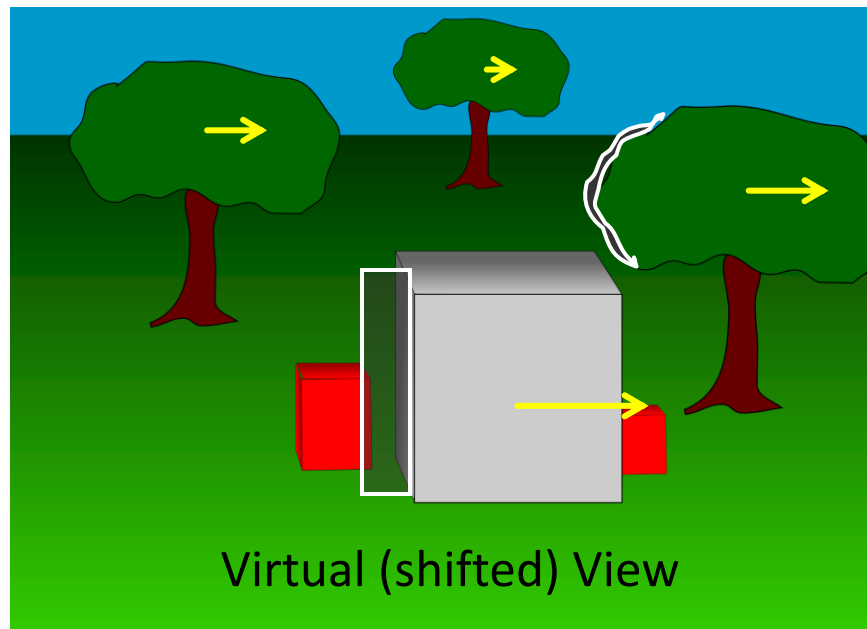
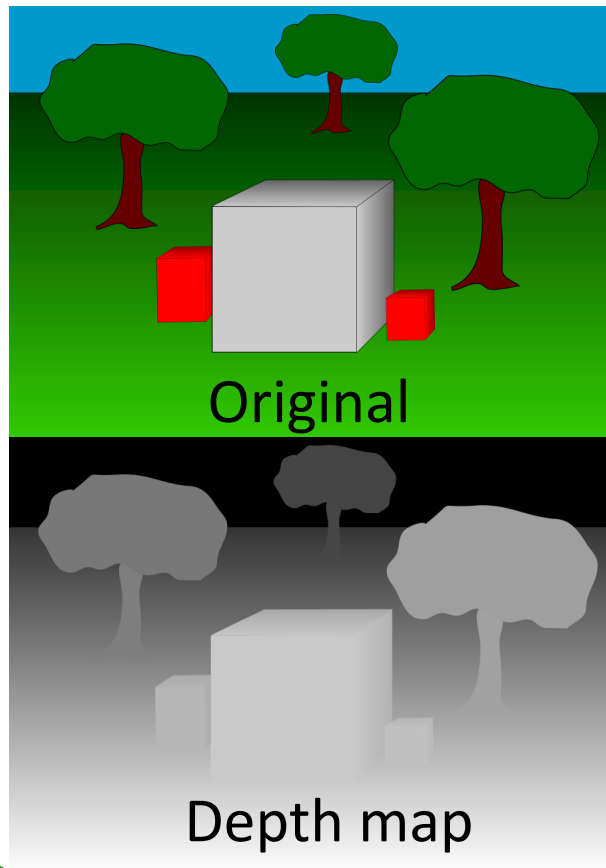




Stereo View Synthesis using Depth-Image-based Rendering (DIBR)

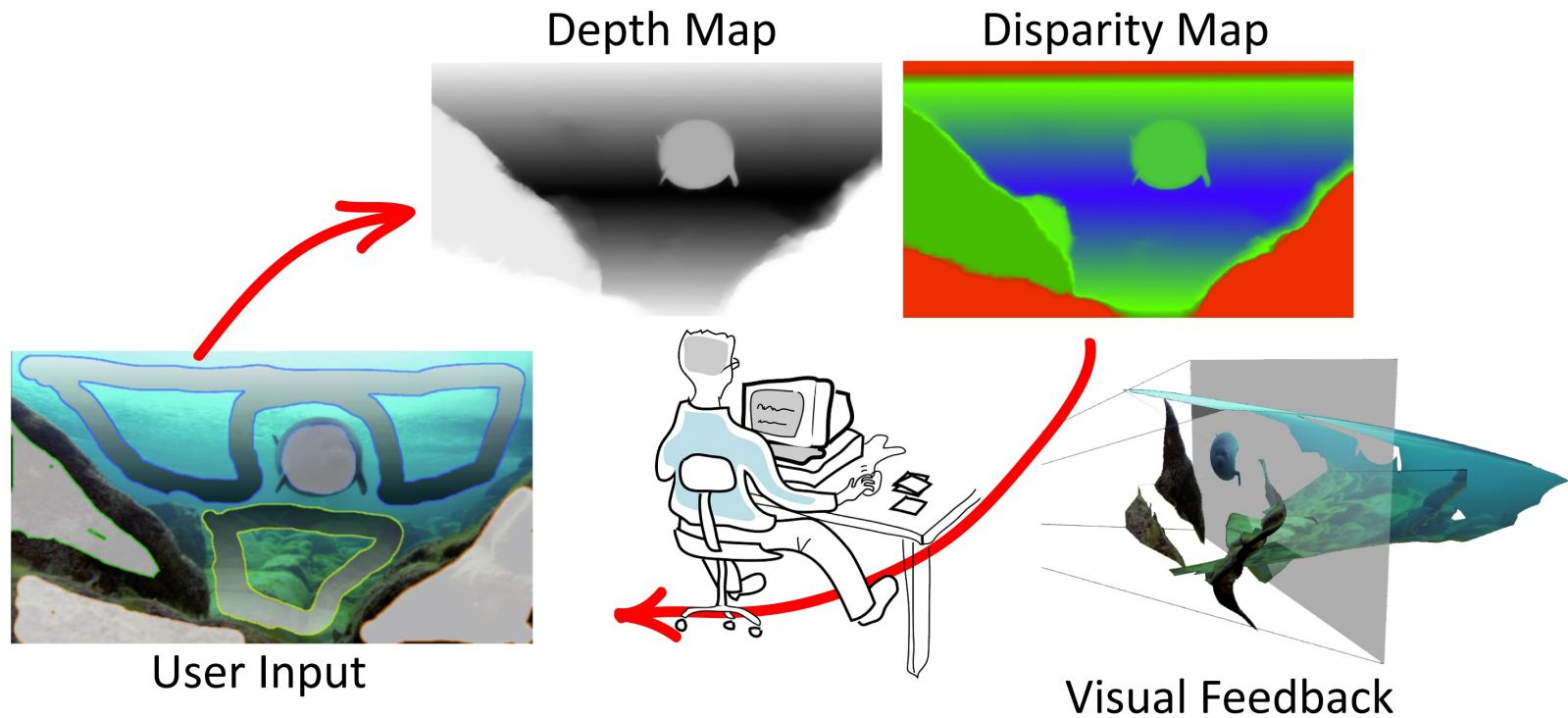


DIBR

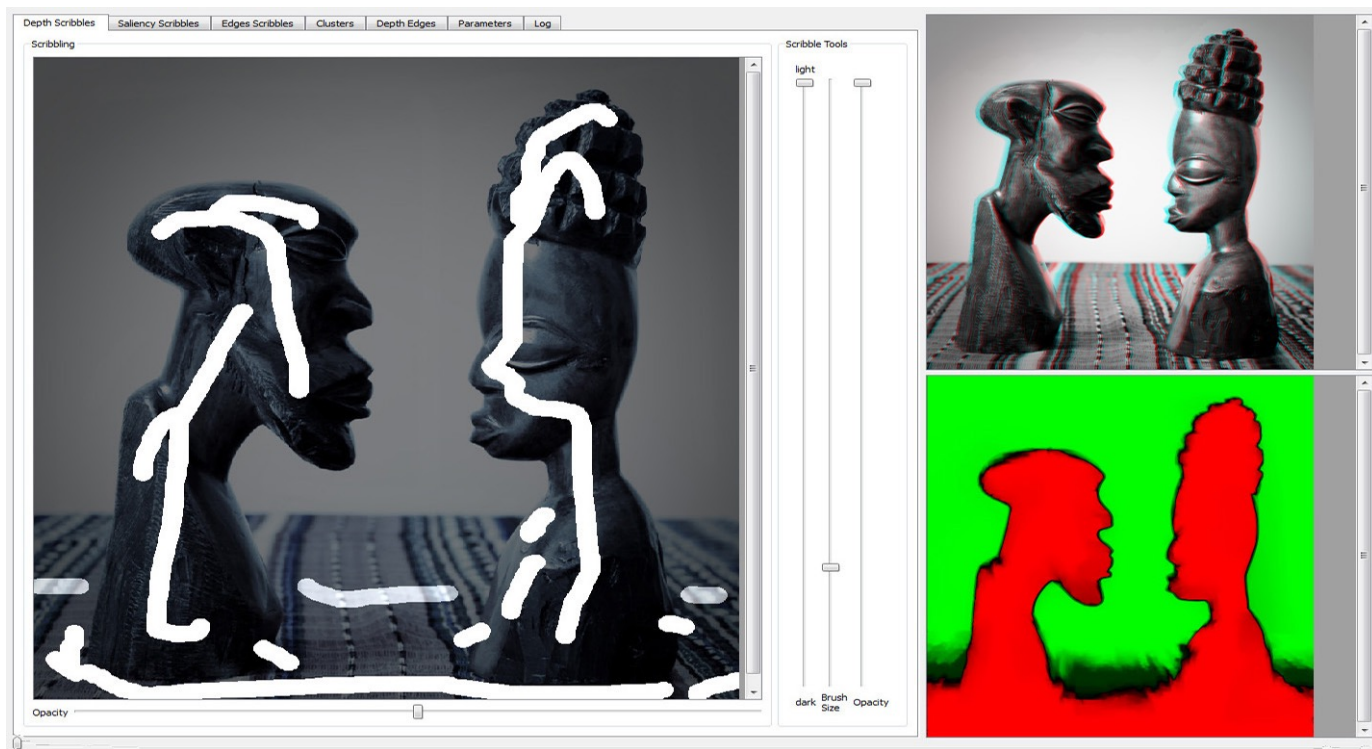


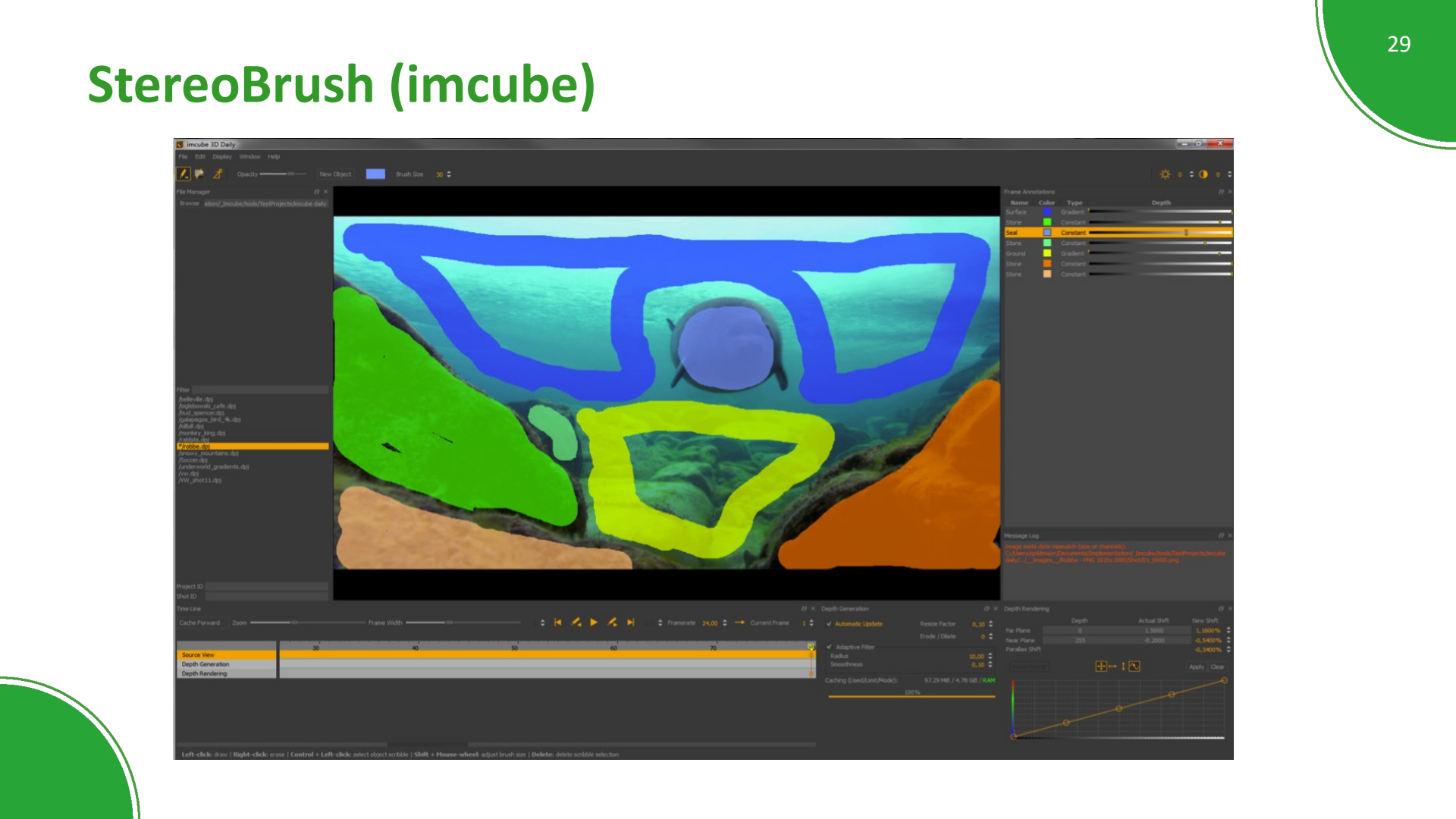
Interactive 2D-to-3D Conversion

StereoBrush

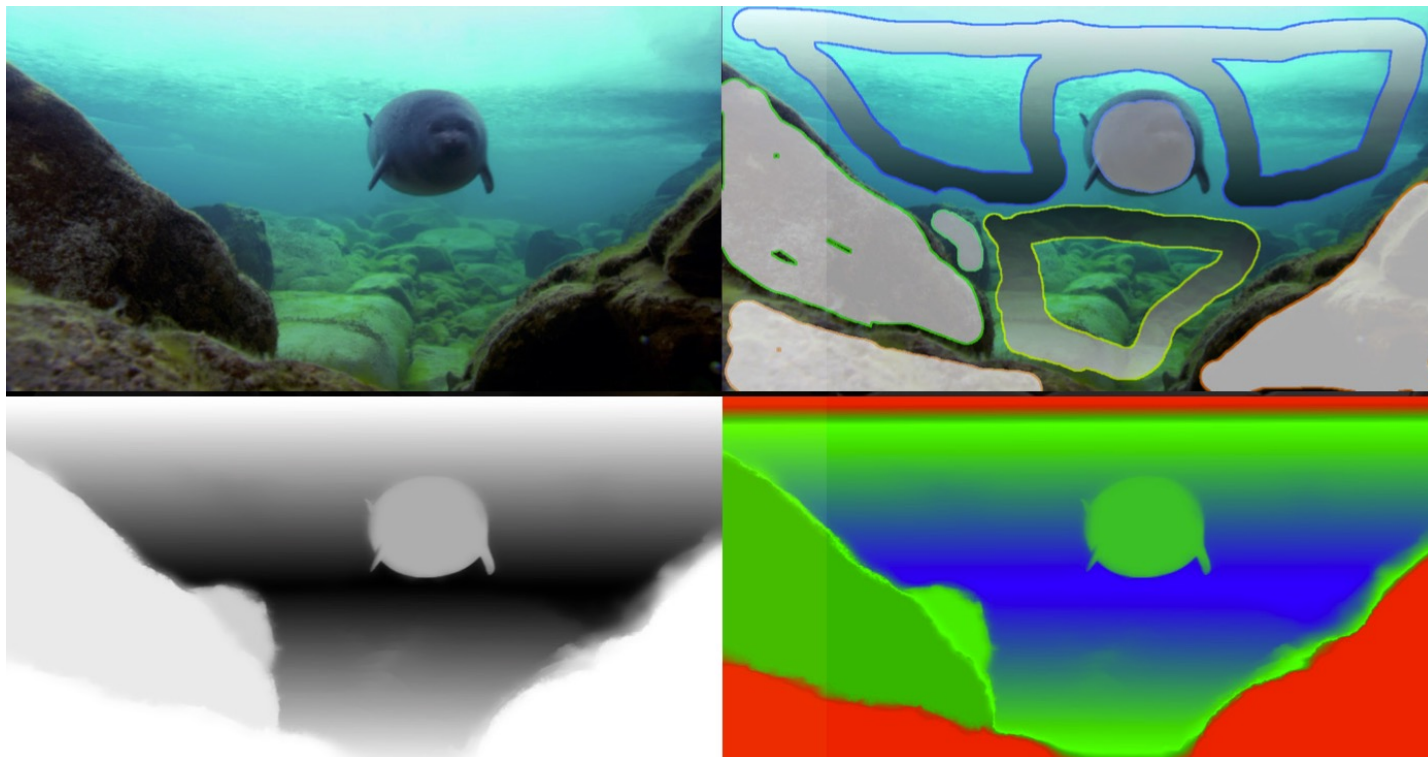


StereoBrush (Disney)

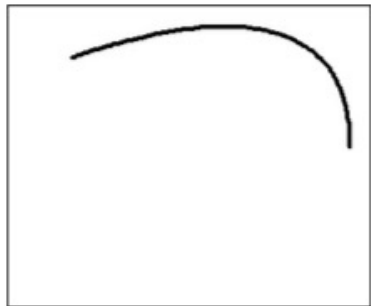




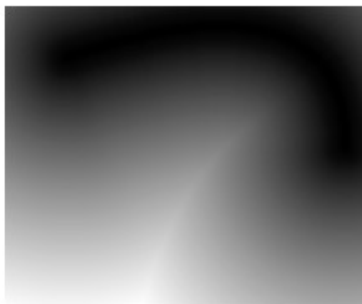
StereoBrush (imcube)



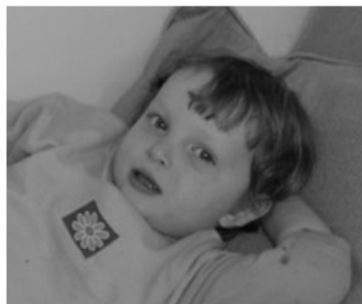
GDT vs. EDT



(a) binary mask M



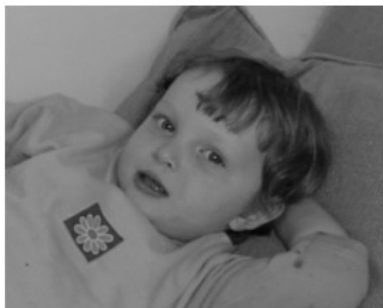
(b) EDT



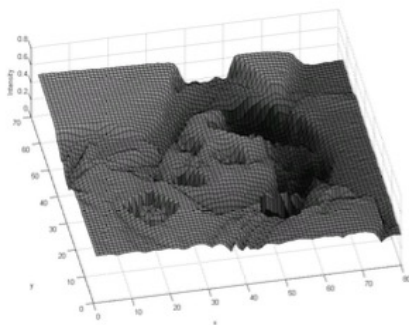
(c) grayscale image I



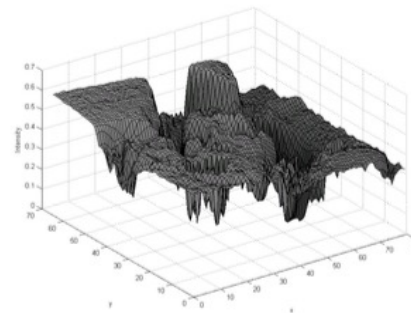
(d) GDT



(a)



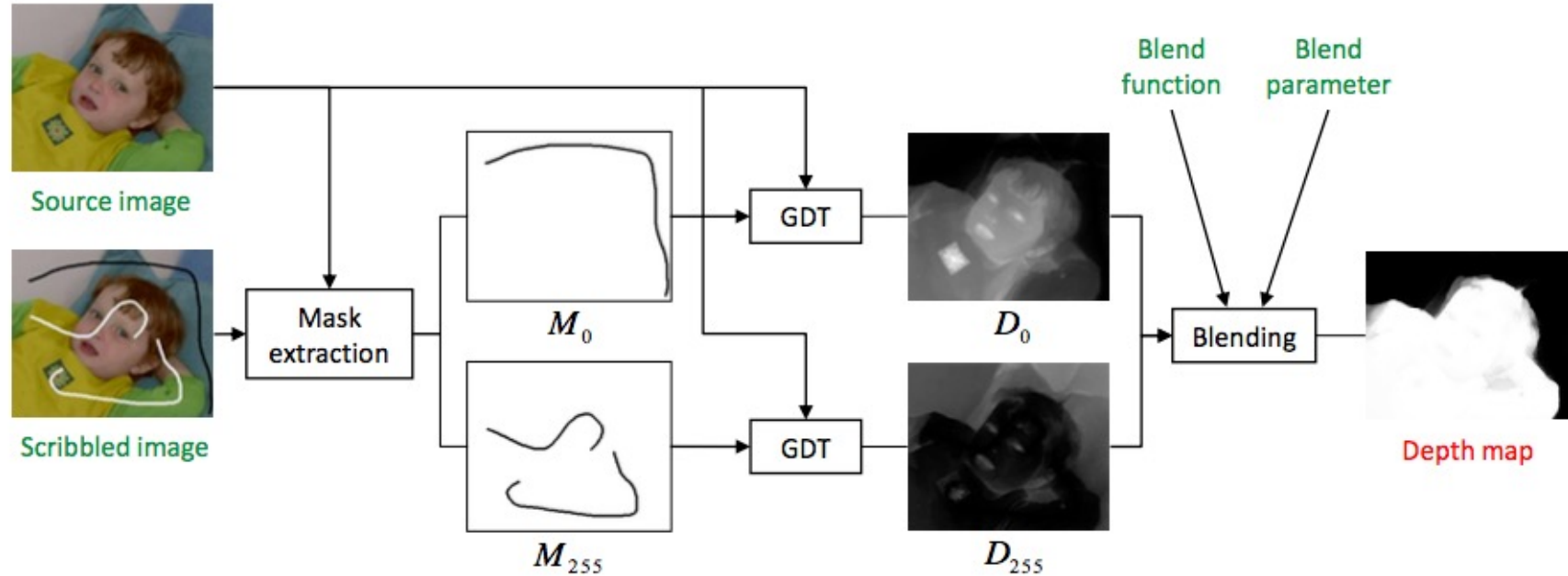
(b)



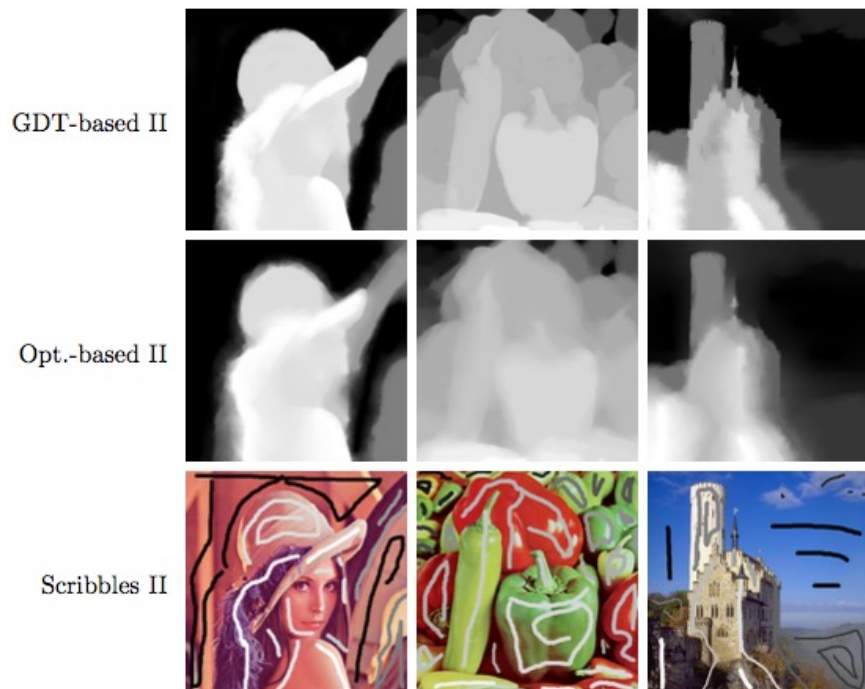
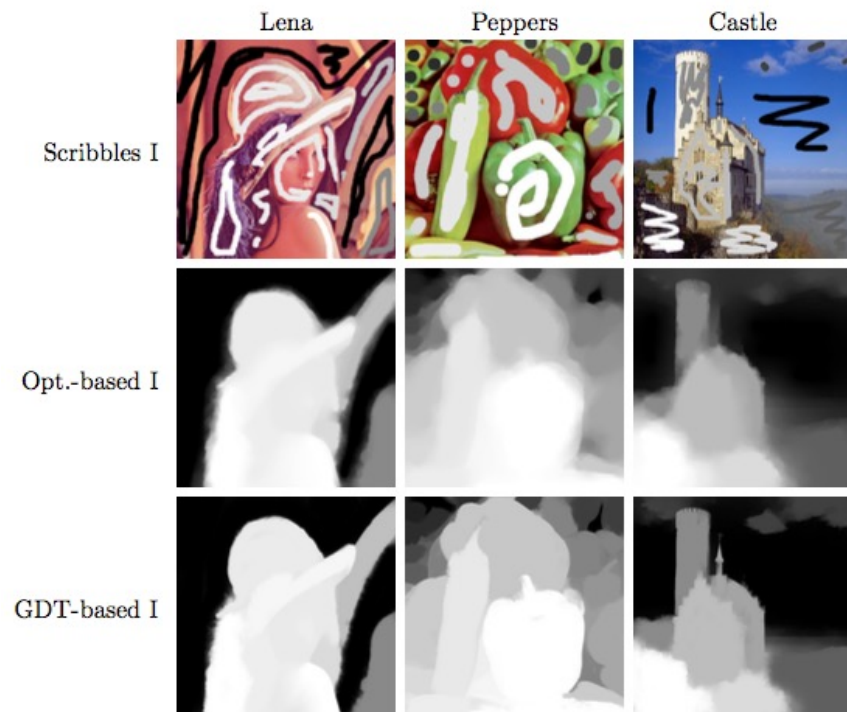
(c)

GDT = geodesic distance transform
EDT = Euclidean distance transform

Geodesic Distance Transform



Comparison



Cinematic 2D-to-3D Conversion

Workflow for Cinematic 2D-to-3D Conversion



3D Storytelling/
Stereographer's
Instructions

- incube 3D Instructor
- incube 3D Daily

- Rotoscoping/ Object Segmentation
- Dimensionalization/ 3D-Compositing
- Inpainting (Occlusion Filling)

- Mocha Pro / Silhouette
- incube 3D Cinema
- Nuke / Photoshop

QC, Stereo
Adjustment and
Finishing

- incube 3D QC Player
- Mistika
- Pablo

Workflow for Cinematic 2D-to-3D Conversion



3D Storytelling/
Stereographer's
Instructions

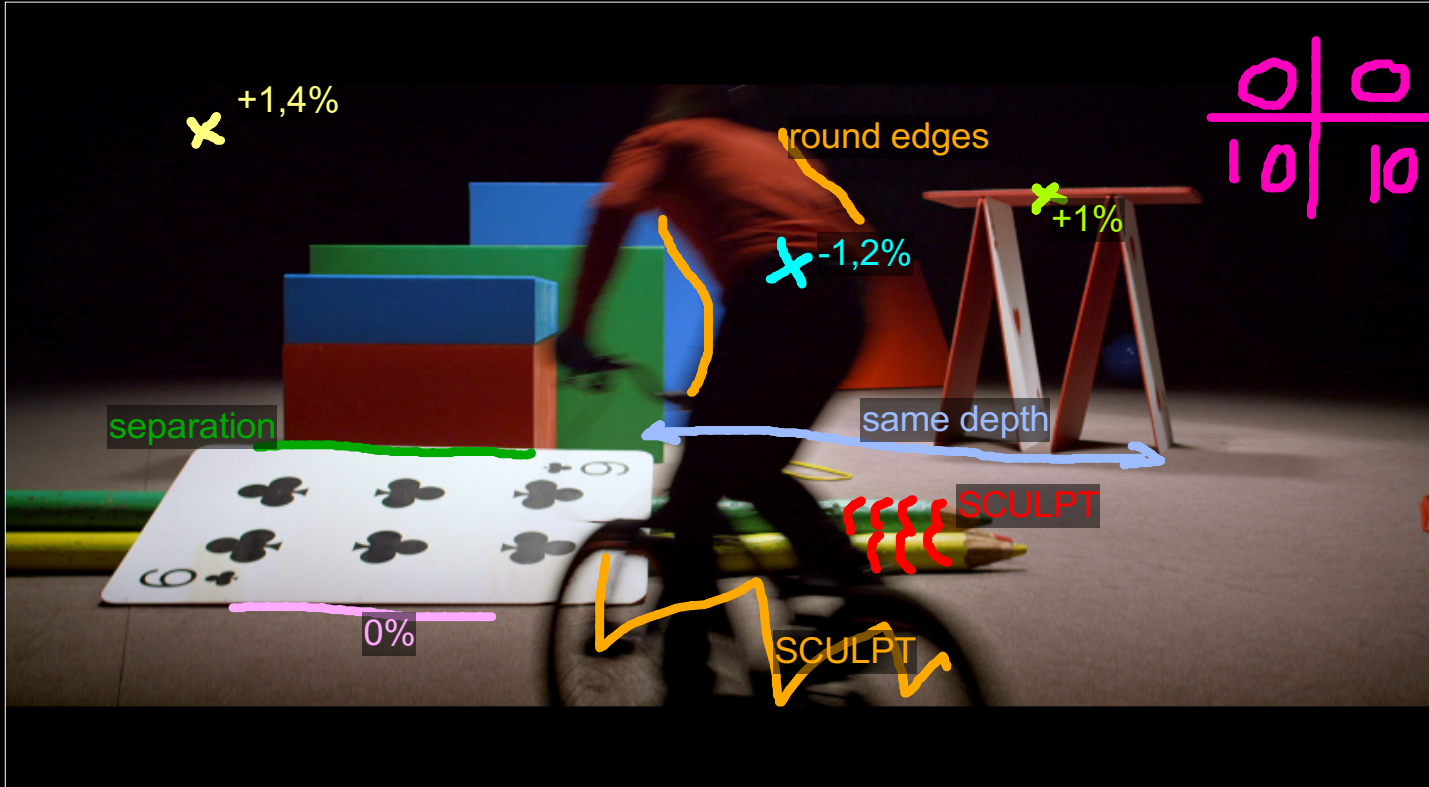
Rotoscoping/ Object Segmentation
Dimensionalization/ 3D-Compositing
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**SCULPT**

Take care of motion blur

Notes:

All boxes need a clear separation between each other and must have an internal volume.

Good 2D to 3D conversion is a creative process!

Workflow for Cinematic 2D-to-3D Conversion



3D Storytelling/
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Instructions

- Rotoscoping/ Object Segmentation
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- Inpainting (Occlusion Filling)

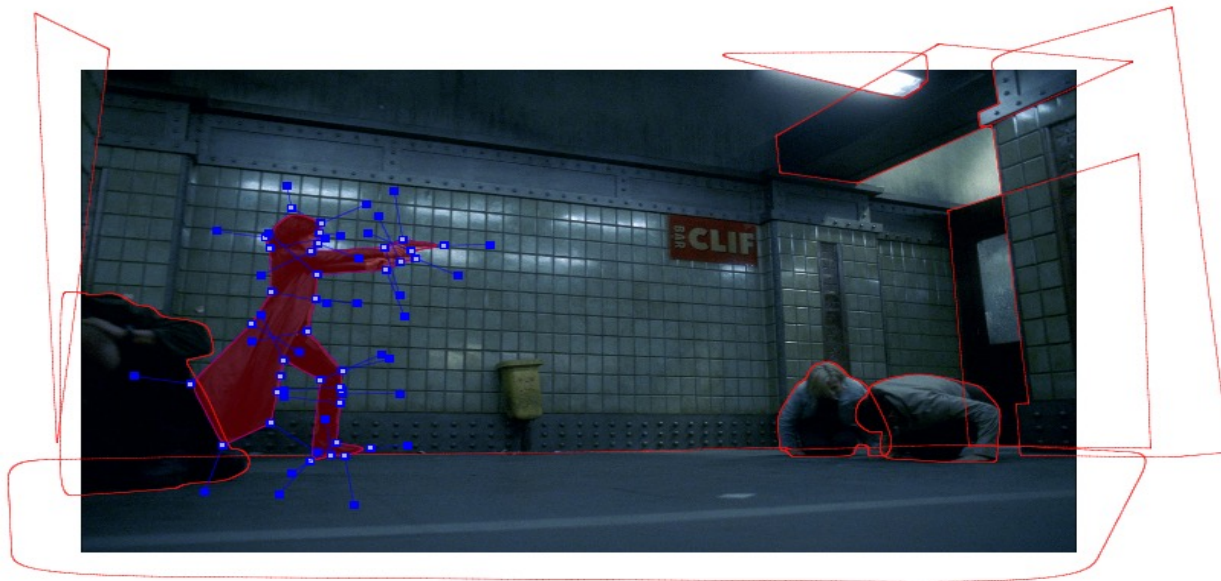
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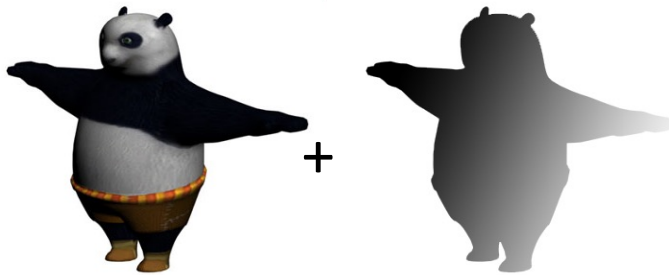
- incube 3D QC Player
- Mistika
- Pablo

Rotoscoping / Object Segmentation

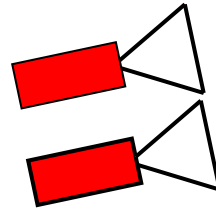


Commercial Tools: Mocha from Imagineer Systems, Adobe After Effects (Roto-Brush), Fusion, Silhouette

Dimensionalization

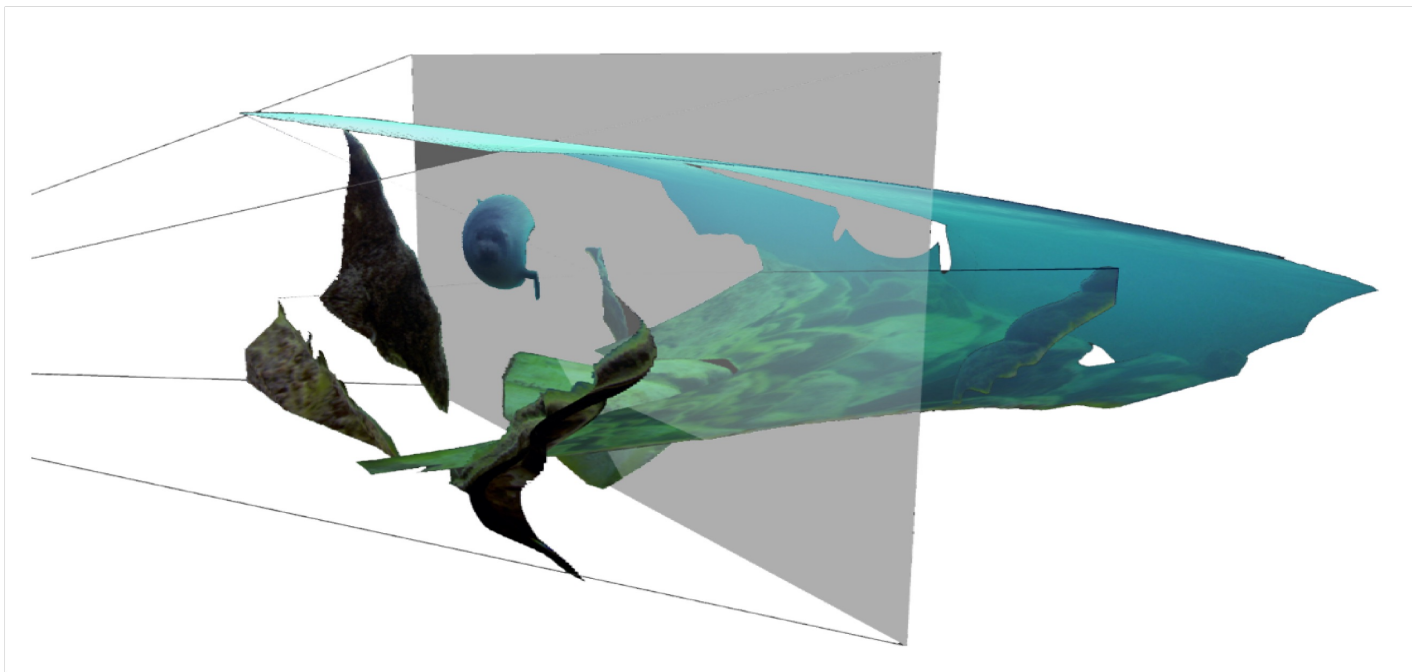


Depth-image-based displacement

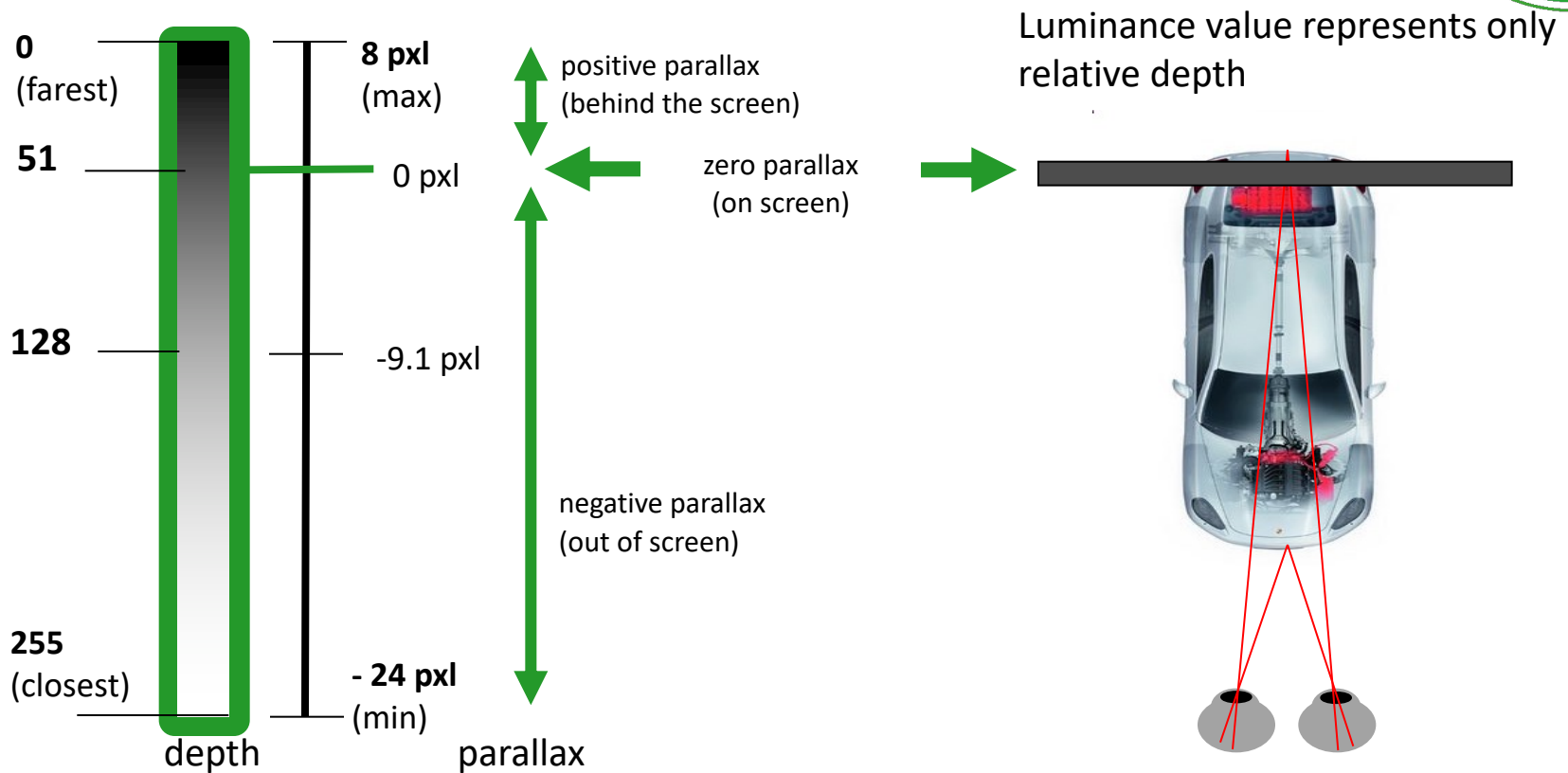


Complex object modeling („Roto-Projection“)

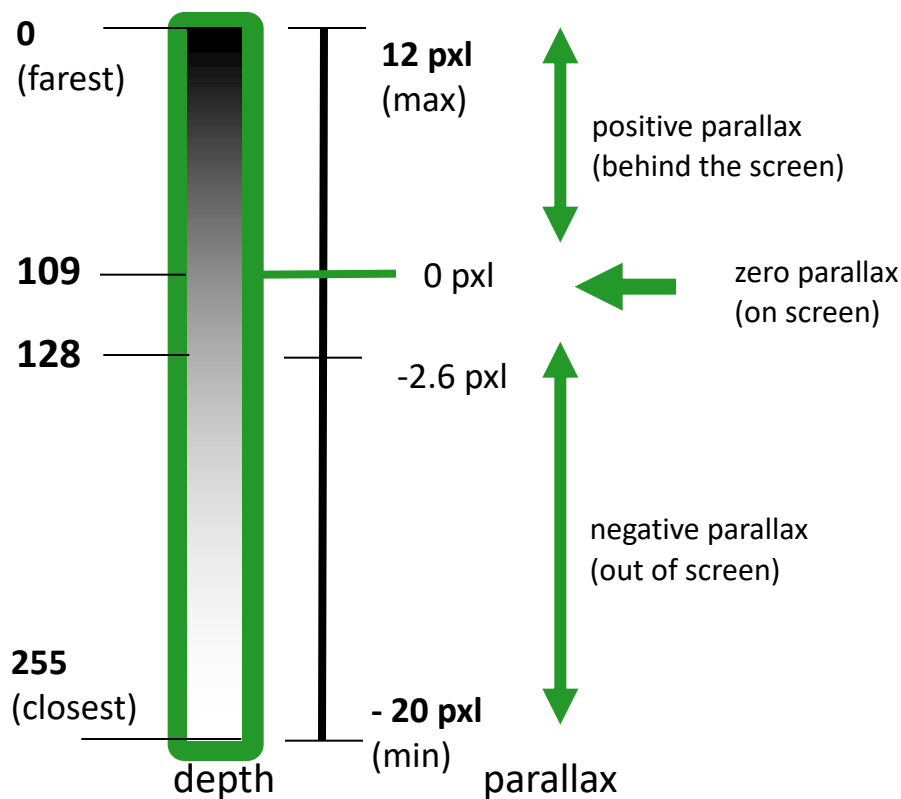
Depth-Image-Based Displacement



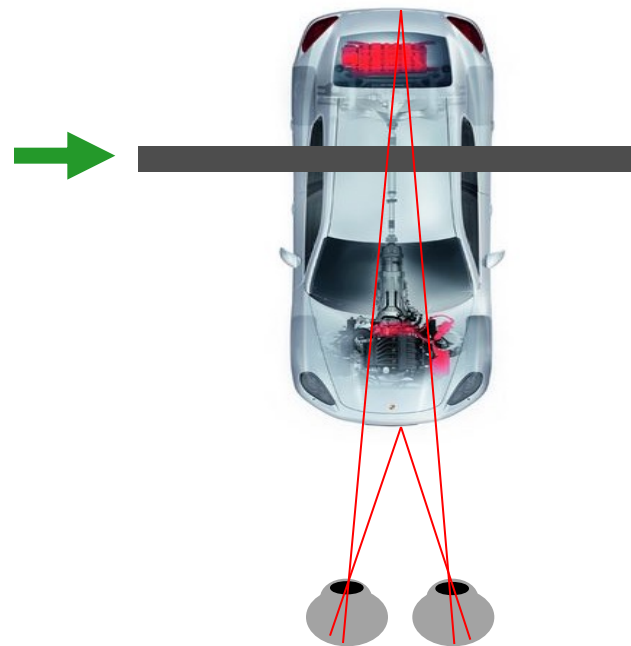
Displacement – Depth Budget



Displacement – Depth Budget



Luminance value represents only relative depth

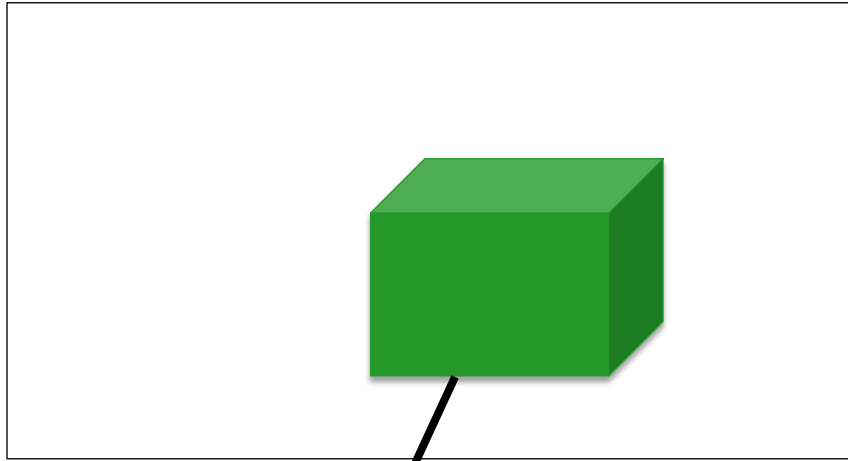




IMCUBE COPY

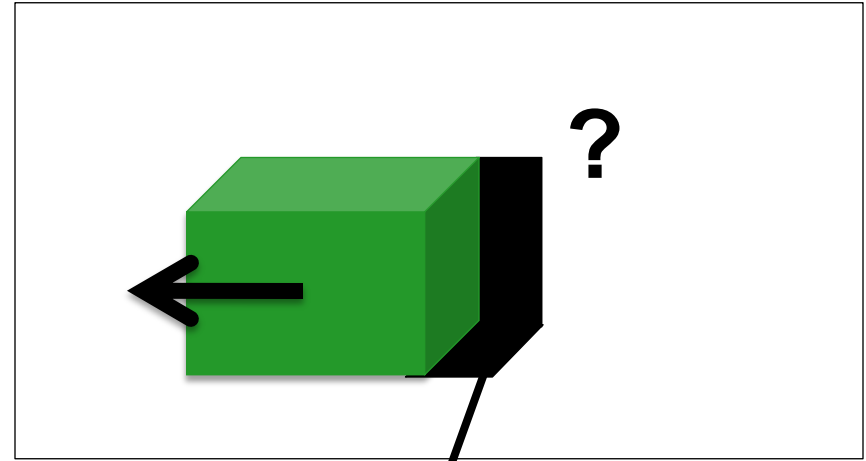
Occlusion Handling

left view



foreground object

right view



disoccluded background

Occlusion Filling / Inpainting



a)



b)



c)

Interpolation vs. Patch-based Inpainting

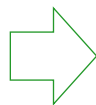


Workflow for Cinematic 2D-to-3D Conversion



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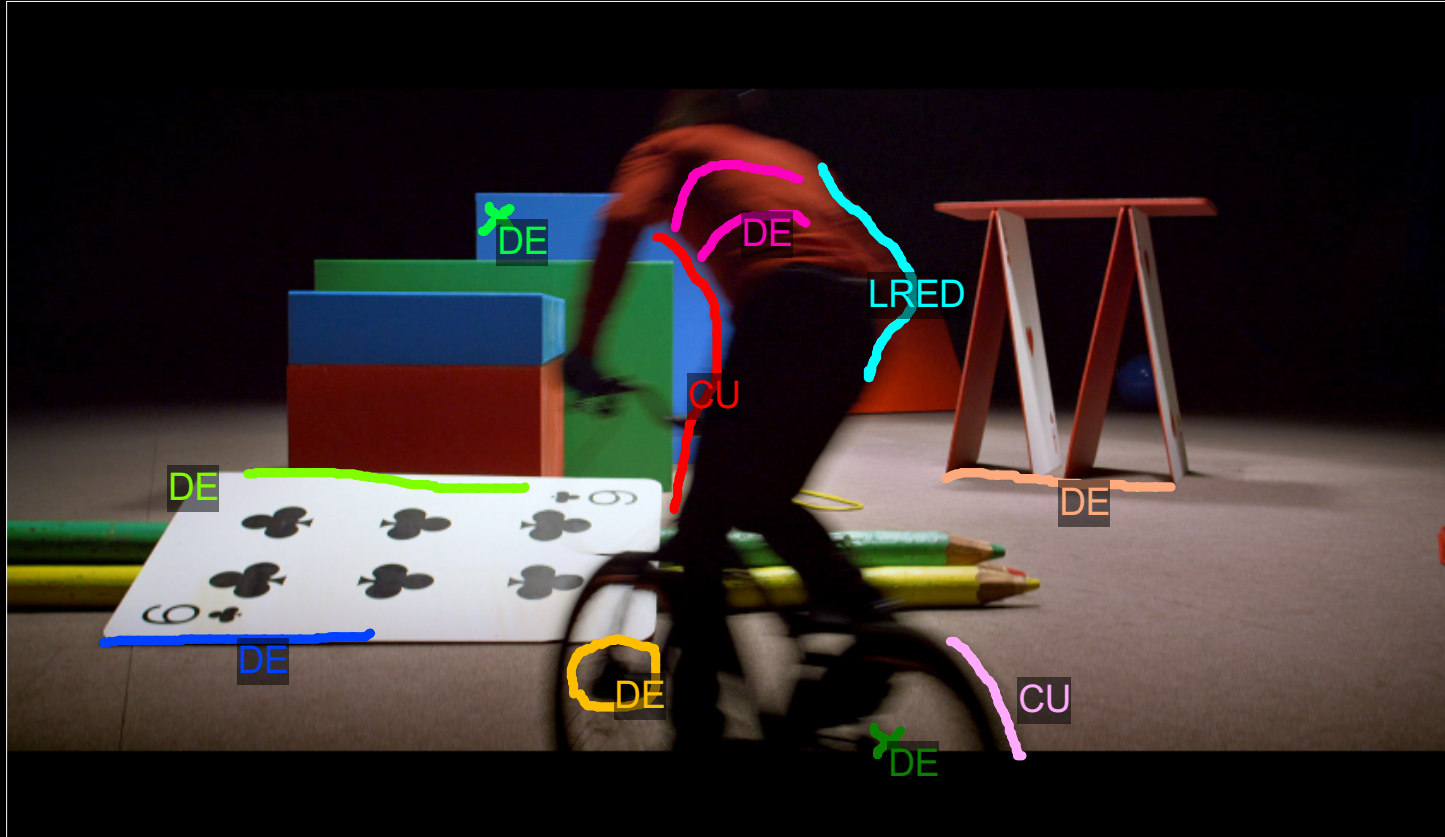
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Adjustment and
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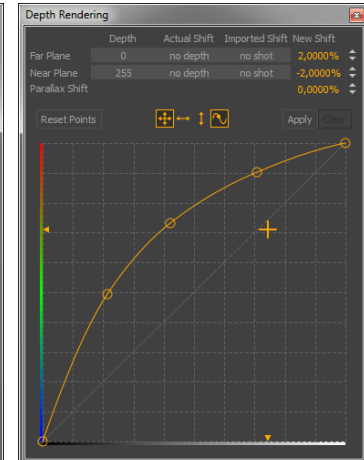
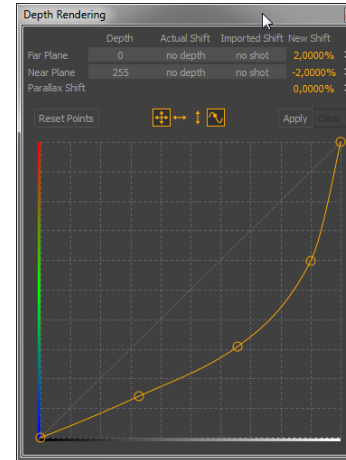
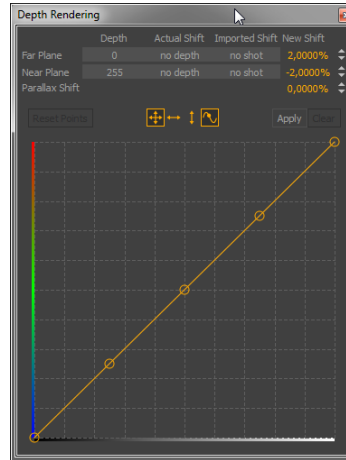




- CU
Glassy edges throughout
- DE
Reduce separation
- DE
Floating
- DE
sculpt
- DE
5px positive
- DE
intersecting ground plane
- DE
Sculpt
- CU
Glassy edges throughout
- DE
4px negative

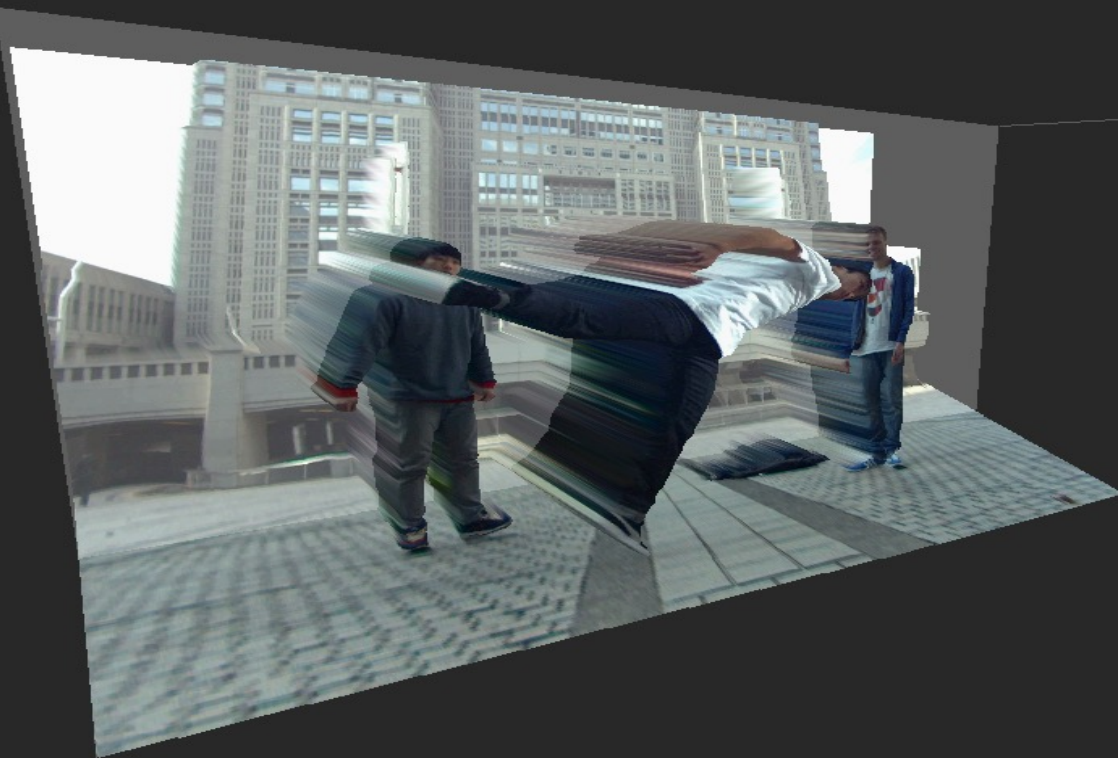
Notes:
next submission in 2K DPX

3D Depth Grading

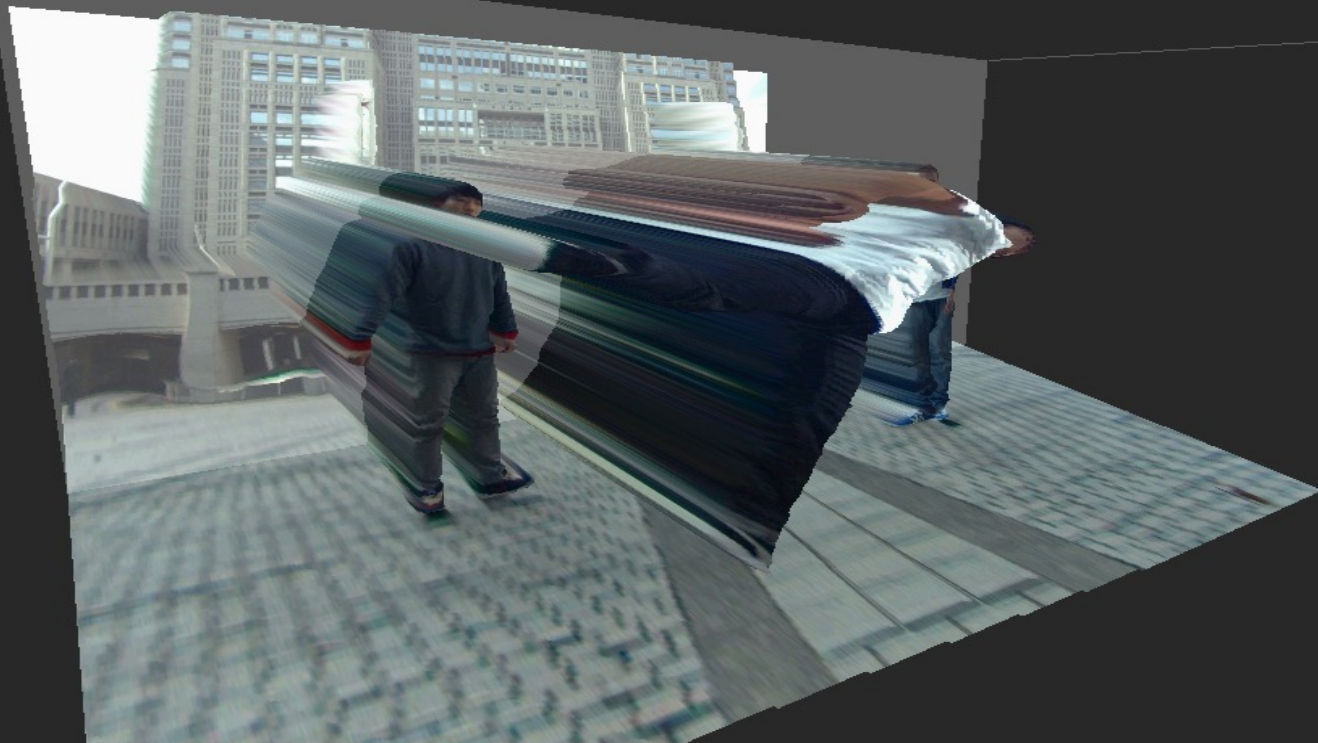


- Depth-grading suite

- Curve Editor for depth-grading/ depth re-mapping







Zusammenfassung

- Was ist natives S3D und welche Probleme gibt es dabei?
- Was ist 2D-3D Konvertierung?
- Was sind die notwendigen Prozesse bei Cinematic 2D-3D-Conversion?



Vielen Dank!

Sebastian Knorr

